Alessandro TEMPIA CALVINO

Senior R&D Staff Engineer

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alessandro-tempia-calvino

My research interests include electronic design automation, logic synthesis, algorithms, new data structures, verification, digital design, and emerging technologies.

Experience

Oct 2024 – pres **Senior R&D Staff Engineer**, *Synopsys Inc.*, Sunnyvale, CA, USA
Research and development of synthesis algorithms for Fusion compiler and Design compiler.

Sep 2020 – Oct 2024 **Doctoral Researcher**, *EPFL (École Polytechnique Fédérale de Lausanne)*, Lausanne, Switzerland
Integrated Systems Laboratory - Electronics design automation (EDA), logic synthesis, and emerging technologies.

Jun 2022 – Oct 2022 **Research Intern**, *X, the Moonshot Factory (Google X)*, Mountain View, CA, USA
Logic synthesis and EDA.

Mar 2020 – Aug 2020 **R&D Engineer**, *Télécom Paris*, EURECOM - Biot, France
Implementation of a model-checker for embedded system models.

Jul 2019 – Dec 2019 **Intern**, *Synopsys*, Montbonnot-Saint-Martin, France
Implementation of timing-driven algorithms for the synthesis of digital circuits.

Education

Sep 2020 – Oct 2024 **Doctor of Philosophy - Ph.D. in Computer and Communication Sciences**, *EPFL (École Polytechnique Fédérale de Lausanne)*, Lausanne, Switzerland
Thesis advisor: Prof. Giovanni De Micheli

Sep 2018 – Sep 2020 **Master of Science in Computer Engineering**, *Télécom Paris*, EURECOM - Biot, France
Specialization in Smart Objects - joint master program with Politecnico di Torino, **GPA 4.0/4**

Sep 2017 – Mar 2020 **Master of Science in Computer Engineering**, *Politecnico di Torino*, Torino, Italy
Specialization in Embedded Systems - joint master program with Télécom Paris, **Full marks with honor (110 cum laude/110)**

Sep 2014 – Sep 2017 **Bachelor's Degree in Computer Engineering**, *Politecnico di Torino*, Torino, Italy

Publications

Conference and Workshop Papers:

- **A. Tempia Calvino**, A. Xess, A. Prasad, J. Minz, E. Testa, W. Lau Neto, A. Mishchenko, G. De Micheli, P. Vuillod, L. Amaru, "Shared Logic Unleashed: Multiple-node Boolean Optimization for Next-Gen Synthesis", accepted, in Design Automation Conference (DAC), 2026.
- F. Marranghello, G. Meuli, B. Gupta, **A. Tempia Calvino**, E. Testa, W. Lau Neto, P. Vuillod, L. Amaru, "Unlocking Automated Datapath Gating via Machine Learning Power Prediction", accepted, in Design Automation Conference (DAC), 2026.
- B. Hien, M. Walter, **A. Tempia Calvino**, A. Mishchenko, R. Wille, "Ashenhurst-Curtis Decomposition Using Don't Cares", in International Workshop on Logic & Synthesis (IWLS), 2025.
- A. Costamagna, **A. Tempia Calvino**, A. Mishchenko, G. De Micheli, "Area-Oriented Optimization After Standard-Cell Mapping", in Asian and South Pacific Design Automation Conference (ASP-DAC), 2025.
- M. Yu, **A. Tempia Calvino**, M. Soeken, G. De Micheli, "Back-end-aware Fault-tolerant Quantum Oracle Synthesis", in Asian and South Pacific Design Automation Conference (ASP-DAC), 2025.
- M. Yu, S. Carpov, **A. Tempia Calvino**, G. De Micheli, "On the Synthesis of High-performance Homomorphic Boolean Circuits", in Workshop on Encrypted Computing & Applied Homomorphic Cryptography (WAHC),

2024.
- S.-Y. Lee, **A. Tempia Calvino**, H. Riener, G. De Micheli, "Late Breaking Results: Majority-Inverter Graph Minimization by Design Space Exploration", in Design Automation Conference (DAC), 2024.
- **A. Tempia Calvino**, G. De Micheli, A. Mishchenko, R. Brayton, "Practical Boolean Decomposition for Delay-driven LUT Mapping", in International Workshop on Logic & Synthesis (IWLS), 2024.
- A. Costamagna, **A. Tempia Calvino**, A. Mishchenko, G. De Micheli, "Post-Mapping Resubstitution For Area-Oriented Optimization", in International Workshop on Logic & Synthesis (IWLS), 2024.
- A. Costamagna, **A. Tempia Calvino**, A. Mishchenko, G. De Micheli, "Area-Oriented Resubstitution For Networks of Look-Up Tables", in International Workshop on Logic & Synthesis (IWLS), 2024.
- **A. Tempia Calvino**, G. De Micheli, "Scalable Logic Rewriting Using Don't Cares", in Design Automation and Test in Europe Conference (DATE), 2024.
- R. Bairamkulov, S.-Y. Lee, **A. Tempia Calvino**, D. Marakkalage, M. Yu, G. De Micheli, "Technology-Aware Logic Synthesis for Superconductive Electronics", in Design Automation and Test in Europe Conference (DATE), 2024.
- **A. Tempia Calvino**, G. De Micheli, "Algebraic and Boolean Methods for SFQ Superconducting Circuits", in Asian and South Pacific Design Automation Conference (ASP-DAC), 2024.
- G. Radi, **A. Tempia Calvino**, G. De Micheli, "In Medio Stat Virtus: Combining Boolean and Pattern Matching", in Asian and South Pacific Design Automation Conference (ASP-DAC), 2024.
- **A. Tempia Calvino**, G. De Micheli, "Technology Mapping Using Multi-output Library Cells", in IEEE/ACM International Conference on Computer-Aided Design (ICCAD), 2023.
- R. Bairamkulov, **A. Tempia Calvino**, G. De Micheli, "Synthesis of SFQ Circuits with Compound Gates", in 2023 IFIP/IEEE 31st International Conference on Very Large Scale Integration (VLSI-SoC), 2023.
- **A. Tempia Calvino**, A. Mishchenko, H. Schmit, E. Mahintorabi, G. De Micheli, X. Xu, "Improving Standard-Cell Design Flow using Factored Form Optimization", in ACM/IEEE Design Automation Conference (DAC), 2023.
- **A. Tempia Calvino**, G. De Micheli, "Technology Mapping Using Multi-output Library Cells", in International Workshop on Logic & Synthesis (IWLS), 2023.
- A. Mishchenko, R. Brayton, **A. Tempia Calvino**, G. De Micheli, "Boolean Decomposition Revisited", in International Workshop on Logic & Synthesis (IWLS), 2023.
- **A. Tempia Calvino**, G. De Micheli, "Depth-Optimal Buffer and Splitter Insertion and Optimization in AQFP Circuits", in Asian and South Pacific Design Automation Conference (ASP-DAC), 2023.
- **A. Tempia Calvino**, G. De Micheli, "Depth-optimal Buffer and Splitter Insertion and Optimization in AQFP Circuits", in International Workshop on Logic & Synthesis (IWLS), 2022.
- G. Meuli, V. Possani, R. Singh, S.-Y. Lee, **A. Tempia Calvino**, D. Marakkalage, P. Vuillod, L. Amarù, S. Chase, J. Kawa, and G. De Micheli, "Majority-based design flow for AQFP superconducting family", in Design Automation and Test in Europe Conference (DATE), 2022.
- L. Apvrille, P. de Saqui-Sannes, O. Hotescu, **A. Tempia Calvino**, "SysML models verification relying on dependency graphs", in International Conference on Model-Driven Engineering and Software Development (MODELSWARD), 2022.
- **A. Tempia Calvino**, H. Riener, S. Rai, G. De Micheli, "A versatile mapping approach for technology mapping and graph optimization", in Asian and South Pacific Design Automation Conference (ASP-DAC), 2022.
- **A. Tempia Calvino**, H. Riener, S. Rai, G. De Micheli, "From logic to gates: A versatile mapping approach to restructure logic", in International Workshop on Logic & Synthesis (IWLS), 2021.
- **A. Tempia Calvino**, L. Apvrille, "Direct model-checking of SysML models", in International Conference on Model-Driven Engineering and Software Development (MODELSWARD), 2021.

Journal Papers and Book Chapters:
- A. Costamagna, **A. Tempia Calvino**, A Mishchenko, G. De Micheli, "Area-Oriented Resubstitution For Networks of Look-Up Tables", in IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2025.
- **A. Tempia Calvino**, G. De Micheli, A Mishchenko, R. Brayton, "Enhancing Delay-driven LUT Mapping with Boolean Decomposition", in IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2024.
- S.-Y. Lee, **A. Tempia Calvino**, G. De Micheli, "Technology Legalization and Optimization for Adiabatic Quantum-Flux Parametron", in IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2024.
- R. Bairamkulov, **A. Tempia Calvino**, G. De Micheli, "Synthesis of SFQ Circuits with Compound Gates", in

VLSI-SoC 2023: Silicon Innovations for Trustworthy Artificial Intelligence, Springer, 2024.
- L. Apvrille, P. Saqui-Sannes, O.A. Hotescu, **A. Tempia Calvino**, "Dependency Graphs to Boost the Verification of SysML Models", in Model-Driven Engineering and Software Development, MODELSWARD 2021-2022, Springer, 2023.
- S. Rai, **A. Tempia Calvino**, H. Riener, G. De Micheli, A. Kumar, "Utilizing XMG-based Synthesis to Preserve Self-duality for RFET-based Circuits", in IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2022.

Preprints:
- M. Soeken, H. Riener, W. Haaswijk, E. Testa, B. Schmitt, G. Meuli, F. Mozafari, S.-Y. Lee, **A. Tempia Calvino,** D. S. Marakkalage, G. De Micheli, "The EPFL logic synthesis libraries", arXiv preprint arXiv:1805.05121v3, 2022.

Patents

- **A. Tempia Calvino**, Xiaoqing Xu, Herman Schmit, "Transistor-level synthesis", US 18462628, 2024.
- Xiaoqing Xu, Herman Schmit, **A. Tempia Calvino**, "Auto-creation of custom standard cells", US 18235437, 2024.

Invited and Conference Talks

Invited Talks:
- **A. Tempia Calvino**, "Technology-aware logic synthesis for superconducting electronics", in International Workshop on Quantum, Cryogenic and Superconductive Computing, Fukuoka, Japan, 2024.
- **A. Tempia Calvino**, "The EPFL Logic Synthesis Libraries: open-source tools for classical and emerging technologies", in Free Silicon Conference (FSiC), Paris, France, 2024.
- **A. Tempia Calvino**, "EPFL Benchmark Results Update", in International Workshop on Logic & Synthesis (IWLS), Zurich, Switzerland, 2024.
- **A. Tempia Calvino**, "Improving Technology Mapping for Standard Cells", at IBM Thomas J. Watson Research Center (online), 2024.
- **A. Tempia Calvino**, "Improving Delay-driven LUT Mapping with Boolean Decomposition", at Efinix Inc. (online), 2024.
- **A. Tempia Calvino**, "Improving Delay-driven LUT Mapping with Boolean Decomposition", at AMD Inc. (Vivado team) (online), 2023.
- **A. Tempia Calvino**, "Technology Mapping Using Multi-output Library Cells", at Google X (online), 2023.
- **A. Tempia Calvino**, "Improving Standard-Cell Design Flow using Factored Form Optimization", at Cadence Design Systems Inc., San Jose (CA), USA, 2023.
- **A. Tempia Calvino**, "EPFL Benchmark Results Update", in International Workshop on Logic & Synthesis (IWLS), Lausanne, Switzerland, 2023.
- **A. Tempia Calvino**, "EPFL Benchmark Results Update", in International Workshop on Logic & Synthesis (IWLS) (online), 2022.
- **A. Tempia Calvino**, "EPFL Benchmark Results Update", in International Workshop on Logic & Synthesis (IWLS) (online), 2021.

Conference Talks:
- **A. Tempia Calvino**, "Practical Boolean Decomposition for Delay-driven LUT Mapping", in International Workshop on Logic & Synthesis (IWLS), Zurich, Switzerland, 2024.
- **A. Tempia Calvino**, "Area-Oriented Resubstitution For Networks of Look-Up Tables", in International Workshop on Logic & Synthesis (IWLS), Zurich, Switzerland, 2024.
- **A. Tempia Calvino**, "Scalable Logic Rewriting Using Don't Cares", in Design Automation and Test in Europe Conference (DATE), Valencia, Spain, 2024.
- **A. Tempia Calvino**, "Algebraic and Boolean Methods for SFQ Superconducting Circuits", in Asian and South Pacific Design Automation Conference (ASP-DAC), Incheon, South Korea, 2024.
- **A. Tempia Calvino**, "In Medio Stat Virtus: Combining Boolean and Pattern Matching", in Asian and South Pacific Design Automation Conference (ASP-DAC), Incheon, South Korea, 2024.
- **A. Tempia Calvino**, "Technology Mapping Using Multi-output Library Cells", in IEEE/ACM International Conference on Computer-Aided Design (ICCAD), San Francisco (CA), USA, 2023.
- **A. Tempia Calvino**, "Improving Standard-Cell Design Flow using Factored Form Optimization", in ACM/IEEE

Design Automation Conference (DAC), San Francisco (CA), USA, 2023.
- **A. Tempia Calvino**, "Technology Mapping Using Multi-output Library Cells", in International Workshop on Logic & Synthesis (IWLS), Lausanne, Switzerland, 2023.
- **A. Tempia Calvino**, "Depth-Optimal Buffer and Splitter Insertion and Optimization in AQFP Circuits", in Asian and South Pacific Design Automation Conference (ASP-DAC), Tokyo, Japan, 2023.
- **A. Tempia Calvino**, "Depth-optimal Buffer and Splitter Insertion and Optimization in AQFP Circuits", in International Workshop on Logic & Synthesis (IWLS), online, 2023.
- **A. Tempia Calvino**, "A versatile mapping approach for technology mapping and graph optimization", in Asian and South Pacific Design Automation Conference (ASP-DAC), online, 2022.
- **A. Tempia Calvino**, "From Logic to Gates: A Versatile Mapping Approach to Restructure Logic", in International Workshop on Logic & Synthesis (IWLS), online, 2021.

Honors and Awards

- **2025 ACM SIGDA Outstanding Ph.D. Dissertation Award in Electronic Design Automation** - *for thesis "Technology Mapping and Optimization Algorithms for Logic Synthesis of Advanced Technologies".*
- O-1 U.S. work visa - *individuals with an extraordinary ability in sciences.*
- Best Student Paper Award at International Workshop on Logic & Synthesis 2024 (IWLS) for the paper "Area-Oriented Resubstitution For Networks of Look-Up Tables".
- Best Student Paper Nomination at International Workshop on Logic & Synthesis 2024 (IWLS) for the paper "Practical Boolean Decomposition for Delay-driven LUT Mapping".
- Best paper award at 2023 IFIP/IEEE 31st International Conference on Very Large Scale Integration System-on-Chip (VLSI-SoC) for the paper "Synthesis of SFQ Circuits with Compound Gates".
- First place in the International Workshop on Logic & Synthesis (IWLS) Contest 2022: "Synthesis of small circuits for completely-specified multi-output Boolean functions represented using truth table".
- Best Student Paper Candidate at International Workshop on Logic & Synthesis 2021 (IWLS) for the paper "From Logic to Gates: A Versatile Mapping Approach to Restructure Logic".
- Best Poster Award at International Conference on Model-Driven Engineering and Software Development (MODELSWARD) 2021 for the paper "Direct Model-checking of SysML Models".
- EDIC I&C Ph.D. Fellowship, EPFL, 2020.

Professional service

Software — Development of *Mockturtle*: an open-source logic synthesis library, *available at*: https://github.com/lsils/mockturtle

Service — Contributing and maintaining open-source software. Maintaining the EPFL logic synthesis libraries, *available at*: https://github.com/lsils/lstools-showcase and contributing to the logic synthesis tool ABC, *available at*: https://github.com/berkeley-abc/abc.

Service — Maintainer of the EPFL Combinational Benchmark Suite and its competition. New updates and best results are presented annually at the International Workshop on Logic & Synthesis (IWLS). *Available at*: https://github.com/lsils/benchmarks.

Reviewer — Reviewed for several top-tier conferences and journals such as TCAD, TODAES, ICCAD, DATE, DAC, IWLS, DDECS, and ISVLSI.

Technical and Personal Skills

Domains of expertise

Logic synthesis, electronic design automation, digital design, microelectronics, algorithms, data structures, computer architectures.

Programming Languages

C, C++, VHDL, Verilog, Java, Python, TCL, Bash, LaTeX, and more.

Languages

- English, Italian, French